

Concretely, we define a normalized parameter space $\alpha \in [-1, 1]^D$ where the origin corresponds to the nominal configuration and the extremes to the hardest limits. The distribution is represented by M virtual particles $\{\mathbf{v}^{(m)}\}_{m=1}^M$, updated through three stages per ADR cycle.

First, a lightweight neural discriminator $D_\phi(\alpha)$ is trained on recent rollout outcomes to estimate the failure probability across the parameter space, providing a differentiable surrogate for the difficulty landscape. Second, the particles are transported via SVGD [18]:

$$\phi^*(\mathbf{v}^{(m)}) = \frac{1}{M} \sum_{j=1}^M \left[\underbrace{k(\mathbf{v}^{(j)}, \mathbf{v}^{(m)}) \nabla_{\mathbf{v}^{(j)}} \log D_\phi(\mathbf{v}^{(j)})}_{\text{driving force}} + \underbrace{\nabla_{\mathbf{v}^{(j)}} k(\mathbf{v}^{(j)}, \mathbf{v}^{(m)})}_{\text{repulsive force}} \right], \quad (3)$$

where $k(\cdot, \cdot)$ is an RBF kernel with bandwidth set by the median heuristic. The **driving force** pushes particles toward high-difficulty regions, while the **repulsive force** prevents mode collapse. Third, the particle cloud is projected into axis-aligned bounds via robust percentile statistics and passed to the simulator.

To prevent catastrophic difficulty escalation, the update is gated by the current success rate: the full VP-SVGd cycle executes only when performance is sufficiently high; when it drops below a low threshold, the particles contract toward the origin until the policy recovers; in between, the distribution is held fixed. This curriculum ensures that the randomization boundary tracks the policy’s evolving competence, as we verify in §4.4.

3.3 Diffusion Policy Distillation

The privileged expert cannot be deployed directly because it requires ground-truth state unavailable in the real world. We distill its behavior into a diffusion policy that takes the frozen geometric features and the robot’s joint positions as input and generates action chunks through iterative denoising. The diffusion formulation is well suited to this setting: domain randomization induces multimodal action distributions that a unimodal policy would collapse by averaging, and predicting action chunks produces temporally coherent trajectories that suppress the oscillations common in contact-rich manipulation.

Data Collection. We roll out the trained expert across the full ADR parameter distribution to collect a dataset $\mathcal{D} = \{(\mathbf{I}_t^h, \mathbf{I}_t^c, \mathbf{q}_t, \mathbf{a}_{t:t+H-1})\}$, where $\mathbf{I}_t^h$ and $\mathbf{I}_t^c$ are the head and chest camera RGB images, $\mathbf{q}_t$ is the robot’s joint position vector, and $\mathbf{a}_{t:t+H-1}$ is a chunk of H consecutive expert actions starting at time t .

Architecture. The diffusion policy contains the frozen geometric encoder (§3.1) as its observation backbone and a 1-D Diffusion Transformer (DiT) [25] as its denoising backbone. We model the conditional distribution over action chunks $\mathbf{a}_0 \in \mathbb{R}^{H \times d_a}$ using a conditional denoising diffusion framework [5, 12]. Each camera image is processed independently by this encoder, and the resulting features are concatenated before entering the DiT through two conditioning pathways.

The dense spatial feature map from the encoder is flattened into a token sequence, and the robot’s joint positions are encoded via a lightweight MLP